

ONATCHIWAY, QC, Canada

WMO: 713870

Lat: 48.894N

Lon: 71.032W

Elev: 997

StdP: 14.17

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.3	-23.2	-36.8	0.7	-29.2	-30.3	1.0	-22.9	17.0	16.8	15.4	14.6	1.6	0	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	82.3	65.1	79.1	63.8	76.1	62.6	69.2	77.0	67.3	74.0	65.6	72.0	6.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	101.7	71.9	64.9	95.7	70.6	63.1	89.7	69.1	34.0	77.0	32.4	74.0	31.0	71.8	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	13.9	11.9	DB	-36.9	88.0	5.9	3.0	-41.1	90.2	-44.6	92.0	-47.9	93.7	-52.2	95.8	(4)
			WB	-37.1	72.5	6.1	1.9	-41.5	73.9	-45.0	75.0	-48.4	76.0	-52.8	77.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.5	3.3	6.6	18.4	33.6	48.5	57.4	62.3	60.4	52.5	40.5	28.3	12.7
	DBStd	22.86	14.96	13.72	13.38	8.96	8.14	7.44	5.64	5.91	7.85	7.67	9.74	13.02
	HDD50	6461	1447	1216	980	492	128	16	0	2	60	310	652	1158
	HDD65	10852	1912	1636	1445	941	514	248	122	168	382	760	1101	1623
	CDD50	1180	0	0	0	1	83	239	380	324	136	16	1	0
	CDD65	95	0	0	0	0	4	21	37	25	8	0	0	0
	CDH74	1101	0	0	0	0	108	320	404	217	51	1	0	0
	CDH80	204	0	0	0	0	21	79	75	23	6	0	0	0
Wind	WSAvg	4.8	4.5	4.7	5.5	6.0	5.2	4.7	4.0	3.9	4.3	4.9	5.2	4.7
Precipitation	PrecAvg	40.0	2.5	2.2	2.2	2.5	3.1	3.8	4.9	4.3	4.4	3.8	3.1	3.1
	PrecMax	46.3	5.7	4.4	4.4	4.2	6.2	5.9	11.5	9.5	8.7	5.7	5.0	4.6
	PrecMin	34.7	0.9	1.2	0.8	1.1	1.2	1.9	2.0	2.0	2.3	1.5	1.0	1.6
	PrecStd	3.2	0.9	0.8	0.9	0.9	1.1	1.1	1.8	1.6	1.3	1.1	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.5	43.0	54.2	63.9	82.8	87.1	86.2	82.8	79.8	68.2	56.8	44.0
		MCWB	41.2	39.4	41.9	47.6	58.9	67.7	67.5	67.5	67.9	57.0	52.5	42.0
	2%	DB	36.0	36.0	45.9	57.2	76.5	81.9	81.9	79.3	74.3	62.5	49.8	37.6
		MCWB	34.8	33.0	38.4	43.7	56.2	63.8	65.8	65.7	64.9	55.8	46.3	36.2
	5%	DB	31.4	32.5	41.1	52.6	71.5	77.4	78.6	76.3	70.0	57.9	45.3	34.1
		MCWB	30.2	30.4	34.9	42.2	54.9	61.2	64.6	64.3	62.2	51.5	42.5	32.8
	10%	DB	25.7	29.0	37.3	48.0	66.1	73.2	75.3	72.9	66.4	53.3	41.1	31.2
		MCWB	24.4	26.9	33.4	39.9	53.2	59.8	63.5	62.9	60.1	48.8	38.9	29.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	40.0	45.0	52.2	64.8	71.1	72.2	70.6	69.8	60.6	52.6	42.1
		MCDB	42.4	41.6	47.9	60.1	73.2	82.7	80.4	78.4	77.7	65.7	55.8	43.3
	2%	WB	35.2	33.3	39.1	47.0	60.6	67.4	69.2	68.4	66.8	56.4	47.4	36.3
		MCDB	36.0	35.3	43.7	53.5	71.2	74.6	76.3	75.5	72.0	61.9	49.4	37.4
	5%	WB	30.4	30.5	36.3	43.3	57.0	64.8	67.1	66.5	63.9	53.0	42.9	33.0
		MCDB	31.3	32.7	40.5	50.5	67.2	73.0	74.2	72.5	68.9	57.0	44.8	33.9
	10%	WB	24.3	26.8	33.2	40.3	54.2	62.3	65.2	64.9	61.2	49.5	39.3	30.1
		MCDB	25.4	29.0	36.9	47.0	64.7	70.2	72.0	70.7	65.8	53.1	41.1	31.2
Mean Daily Temperature Range	5% DB	MDBR	22.1	24.7	24.2	20.7	24.5	24.0	21.9	21.0	20.2	15.0	12.9	17.2
		MCDBR	20.1	21.3	26.7	29.3	36.7	31.7	28.2	26.5	25.0	24.7	17.7	16.0
		MCWBR	19.7	18.7	19.5	18.3	18.8	15.7	14.2	14.2	15.6	17.1	14.5	15.1
	5% WB	MCDBR	20.7	19.7	21.5	25.2	27.2	24.8	21.0	20.2	20.9	21.2	15.8	15.9
		MCWBR	20.5	17.8	16.3	17.2	15.9	13.9	11.8	11.7	14.0	16.1	14.2	15.6
Clear-Sky Solar Irradiance	taub	0.257	0.276	0.302	0.365	0.347	0.375	0.390	0.372	0.359	0.320	0.303	0.255	
	taud	2.281	2.234	2.211	2.222	2.438	2.385	2.361	2.414	2.434	2.531	2.418	2.338	
	Ebn at Noon	257	278	288	279	290	281	274	275	267	262	238	243	
	Edh at Noon	23	30	37	41	34	36	37	34	30	23	20	19	
All-Sky Solar Radiation	RadAvg	398	714	1112	1425	1648	1750	1630	1371	1043	596	320	254	
	RadStd	53	91	101	114	154	123	153	87	78	62	31	28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		-1.37	N/A	N/A	N/A	N/A	N/A	+443	+472	N/A	N/A
Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page